Hybrid Deep Learning-Based Intrusion Detection for Cyber-Physical Systems with Explainability

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Abstract

Cyber-Physical Systems (CPS) are increasingly vulnerable to cyberattacks due to their interconnected nature in Industrial Internet of Things (IIoT) environments. Detecting such attacks is challenging due to high-dimensional and complex network traffic data. This study proposes a hybrid deep learning-based intrusion detection system combining Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM), and Generative Adversarial Networks (GAN) for enhanced detection performance. The model is evaluated using the Edge-IIoT Balanced Subset dataset containing 50,000 samples across multiple attack categories. After preprocessing, a total of 14 attributes were retained from the dataset. Out of these, 13 were used as input features for model training, while the remaining attribute (Attack_Label) was used as the target variable. Features such as Frame_Time, Source_Port, and Destination_Port were excluded as they do not significantly contribute to intrusion detection and may introduce noise into the model. CNN is used for spatial feature extraction, LSTM captures temporal dependencies, and GAN improves data balance through synthetic sample generation. Additionally, Explainable Artificial Intelligence (XAI) techniques are applied to interpret model decisions. Experimental results show that the proposed model achieves an accuracy of 96.2%, demonstrating its effectiveness in detecting cyberattacks while maintaining interpretability. This approach provides a reliable and scalable solution for CPS security.

Keywords—Cyber-Physical Systems, Intrusion Detection System, CNN- LSTM, Generative Adversarial Networks, Explainable Artificial Intelligence, Network Traffic Analysis, Attack Detection, Edge- IIoT Balanced Subset for Intrusion Detection Dataset.

I. INTRODUCTION

The rapid expansion of intelligent industrial infrastructure has led to increased usage of Cyber-Physical Systems (CPS) across numerous modern applications, including Smart Manufacturing, Intelligent, Smart Grids and robots [5], [7]. CPS combines computational intelligence, communication networks, and physical processes to monitor in real-time, exchange data, and provide automated control over physical environments via the Industrial Internet of Things (IIoT) [10], [22]. While this level of integration provides enhanced operational efficiency and automation, the level of interconnectivity within CPS also creates new, unique challenges with regard to cybersecurity [6], [7]. Due to their interconnectivity, CPS environments have become primary targets for sophisticated cyber-attacks that target various vulnerabilities in both network communication and the interaction of devices with each other [5], [20].

Traditional intrusion detection systems typically rely on a combination of signature or rule-based mechanisms for detection; thus, these mechanisms are not designed to detect attacks that have evolved over time and are complex when looking across multiple CPS networks [14], [17]. Also, the volume of high-dimensional network traffic generated by CPS environments makes the task of detecting attacks significantly more difficult using conventional machine learning methods [18].

In recent years, deep learning techniques have shown great promise for use in cybersecurity applications because they automatically extract complicated patterns from very large datasets [13], [19]. To address this problem, we proposed a hybrid CNN-LSTM based intrusion detection system, along with an enhanced version that incorporated GAN (Generative Adversarial Network) based data augmentation techniques, as well as a hybrid explainable AI (XAI) model that is able to produce interpretable results. We performed the evaluation for the newly developed hybrid models using the Edge-IIoT Balanced Subset for Intrusion Detection dataset. It contains 50,000 samples for five traffic classes (DDoS, DoS, MITM, Injection and normal traffic) and an improvement in intrusion detection accuracy, while providing interpretable outputs of the top features that influence the security of CPS networks.

Despite significant advancements in deep learning-based intrusion detection systems, several critical challenges remain unresolved. Most existing approaches focus primarily on improving detection accuracy while neglecting model interpretability, which is essential for decision-making in critical cyber-physical systems. Additionally, many methods fail to effectively handle class imbalance in network traffic data and do not integrate spatial, temporal, and data augmentation techniques within a unified framework. Furthermore, existing models often lack transparency, making it difficult for security analysts to trust automated decisions. These limitations highlight the need for a hybrid, explainable, and robust intrusion detection system for CPS environments. The main contribution of this work is the integration of CNN, LSTM, and GAN within a unified framework along with explainable AI techniques to improve both detection accuracy and model transparency in CPS environments.

II. LITERATURE REVIEW

A. CNN-LSTM Based Intrusion Detection

CNNs have become one of the most commonly used methods of feature extraction in cybersecurity applications [13], [14]. CNNs can effectively model the spatial relationships among input features and automatically learn hierarchical representations from network traffic data, thereby improving the classification accuracy of intrusion detection [14]. LSTMs are also used extensively for time series data analysis, including traffic flows [2], [19]. In network security applications, LSTMs have been successfully applied to detect temporal relationships in traffic flows, as well as to recognize abnormal behaviors that develop over time [2], [19]. LSTM models capture sequential dependencies, while CNN models capture spatial dependencies.

B. Generative Adversarial Network Based Augmentation

A major challenge faced by intrusion detection systems is class imbalance, where certain types of attacks have significantly fewer samples compared to the large amount of normal network traffic [1], [5]. This imbalance can negatively affect the performance of deep learning models. To address this problem, researchers have explored the use of Generative Adversarial Networks (GANs) for data augmentation [1], [5]. GANs consist of two competing neural networks—a generator and a discriminator—that work against each other during training to produce realistic synthetic data samples. In cybersecurity, GAN-based and similar models are employed to generate additional attack samples, helping to balance datasets and improve the robustness of intrusion detection systems [1], [5]. By augmenting the training data with these synthetic samples, GANs enable deep learning models to better represent rare attack patterns and learn them more effectively.

C. Explainable Artificial Intelligence (XAI) in cybersecurity

Many deep learning models have experienced great success when used in security but many of these deep learning models are essentially "black box" in nature and thus have a major limitation with regards to how the prediction was made (i.e., not easily interpretable) [13], [18]. As no explanation exists for the prediction made; it creates a negative impact on how much trust the user placed upon utilizing an Intrusion Detection System (IDS) that is automated, because the prediction lacked sufficient transparency. In addition, if users have little or no trust in utilizing an IDS - Automated, then they may not incorporate automated methods into critical Infrastructures (i.e., cyber/physical systems), thereby limiting the practical applicability [18]. These methods provide insight into how / why a model made a particular decision by providing an understanding of which features and/or internal components within the model contains the greatest impact or influence on the final classification outcome. Using explainability techniques in an Intrusion Detection (ID) system provides security analysts with an understanding of which characteristics are present in network traffic that help to identify malicious activity [13], [18]. Researchers have used methods such as feature importance analysis, LIME (layer-wise interpretation), and contribution analysis to determine the influence of different model features as well as how specific neural network layers impact the overall classification outcomes. Therefore, by applying explainable Artificial Intelligence (XAI) methods to deep learning models, Intrusion Detection Systems provide both detection performance and meaningful explanations of their predictions of detected incidents.

D. Comparative Analysis of existing approaches

With the increasing reliance on Cyber-Physical Systems, cybersecurity and intrusion detection have become major concerns [5], [7]. Researchers have explored the use of AI and deep learning to improve detection system performance, make systems more resilient to failures, and reduce cyber threats [13], [19]. Techniques like CNN, LSTM, and GAN have been studied extensively to enhance intrusion detection [1], [14], [19]. These approaches have proven to be much better at detecting complex and new attack patterns compared to traditional methods [14], [17]. Table 1 gives an overview of recent research in intrusion detection systems, covering different datasets, methods, key contributions, and limitations.

Ref	Author & Year	Technique Used	Dataset / Application Area	Key Strength
[2]	Levytskyi et al. (2024)	Deep Learning + Reinforcement Learning	Autonomous CPS	Improves adaptive decision-making
[3]	Divya et al. (2025)	Hybrid RL + Supervised Learning	CPS fault detection	Enhances fault tolerance
[5]	Sharma (2024)	CPS Security & Privacy Framework	Cyber-Physical Systems	Addresses security challenges
[6]	Phillips et al. (2024)	Knowledge-based Risk Assessment	CPS systems	Automated cybersecurity risk analysis
[7]	Perone et al. (2024)	Cybersecurity Risk Assessment	Autonomous CPS	Focus on safety and security
[13]	Shelke et al. (2025)	Federated Learning for CPS Security	CPS security datasets	Decentralized intrusion detection

Table 1: Comparison between existing techniques

The majority of the studies reviewed provide evidence that AI and Deep Learning have improved Intrusion Detection and cybersecurity performance [13], [19]. However, there continue to be several areas of concern that must be addressed relating to computational complexity, limited scalability of models, and lack of interpretability [6], [18], [21]. These areas of concern highlight the importance of developing hybrid systems which combine computationally efficient detection methods with greater transparency and explainability than currently available.

E. Research Gap

Despite extensive research on deep learning techniques such as CNNs, LSTMs, and GANs for intrusion detection in cyber-physical systems, most existing approaches primarily emphasize detection accuracy while overlooking critical aspects of model explainability and interpretability [8], [11], [27]. Several studies have explored deep learning–based intrusion detection frameworks and highlighted challenges such as data imbalance, spatiotemporal feature extraction, and model complexity; however, transparency in model decision-making remains insufficiently addressed [26], [28].

Although GANs have been investigated to mitigate issues such as class imbalance and the scarcity of attack samples, their integration with hybrid deep learning architectures is still limited [27], [17]. Furthermore, existing research in cyber-physical system security largely focuses on enhancing system robustness, risk assessment, and AI-driven threat detection, with comparatively less attention given to interpretable model design [5], [6], [13].

In addition, only a few studies adopt a bottom-up interpretability perspective, where model behavior is analyzed through feature contribution assessment and internal layer-wise understanding [18], [21]. This lack of interpretability reduces the transparency, trustworthiness, and practical deployment of intrusion detection systems in safety-critical cyber-physical environments [4], [7].

To address these gaps, this research proposes a hybrid deep learning-based intrusion detection system that integrates a CNN–LSTM architecture with GAN-based data augmentation, while explicitly incorporating explainability techniques. These include layer-wise interpretability analysis and feature importance assessment to identify the most influential features and internal network components contributing to final predictions. By doing so, the study aims to enhance both detection performance and the transparency of intrusion detection systems in cyber-physical systems.

III. METHODS

A. System Architecture

This proposed intrusion detection framework contains multiple sequential phases such as data preprocessing, feature extraction, classification, attack detection, and explanation [3], [15]. The Cyber-Physical Systems (CPS) are combination of physical components, cyber components, and communication components [5], [9].

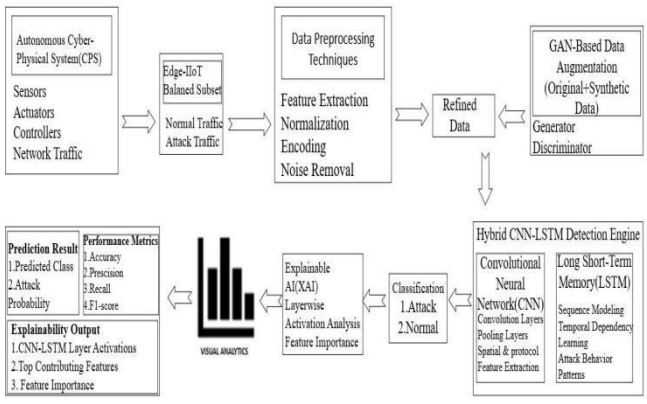


Fig. 1. System Architecture

Initially, network traffic data from the dataset was preprocessed to remove inconsistencies and normalize feature values. Next, preprocessed data was provided as input to a hybrid deep learning architecture with combined CNN and LSTM layers [3], [15]. The CNN portion contains CNN layers that perform the automatic extraction of features by identifying spatial relationships within input features, and the LSTMs learn the temporal dependencies associated with sequential network traffic flows [2], [19]. To enhance the overall robustness of the model, synthetic data samples were generated via a GAN during the training phase of the framework [1], [5]. XAI techniques were also used to analyze predicted attacks based on the model's predictions to determine which features were most influential in detecting cyberattacks [13], [18].

B. Dataset Description

The Edge-IIoT Balanced Subset dataset was selected due to its relevance to real-world Industrial Internet of Things (IIoT) environments. It contains approximately 50,000 network traffic samples, including both normal and malicious activities. The dataset includes multiple attack categories such as Distributed Denial of Service (DDoS), Denial of Service (DoS), Man-in-the-Middle (MITM), and Injection attacks.

Initially, the dataset contained 17 attributes. After preprocessing, 14 attributes were retained. Among these, 13 attributes were used as input features for model training, while one attribute (Attack_Label) was used as the target variable.

Features such as Frame_Time, Source_Port, and Destination_Port were removed during preprocessing, as they do not significantly contribute to intrusion detection and may introduce noise into the model. We used 70% of the dataset for training and 30% of the dataset for testing.

The balanced nature of the dataset ensures that the model does not suffer from bias toward any particular class, thereby improving detection performance and generalization.

C. Data Preprocessing

The first step is to clean the data by removing any rows with missing values or columns with irrelevant features to ensure we have quality data to work with. After cleaning and preparing your data, a feature scaling technique must be used to convert all numeric values into a common range (i.e., between 0 and 1). This allowed your neural networks to have more significant stability and converge more rapidly during training [18]. Feature selection helps in reducing model complexity, improving training efficiency, and enhancing detection accuracy by eliminating redundant and irrelevant attributes.

Normalize Equation:

$$X_{norm} = (X - X_{min}) / (X_{max} - X_{min})$$

Where;

X = original feature value

Xmin = minimum feature in the dataset

Xmax = maximum feature in the dataset

Xnorm = normalized value

D. CNN-LSTM Intrusion Detection Model

The primary part of the recommended framework is a combined CNN and LSTM deep learning network for recognizing cyberattacks based on information acquired from network traffic data [3], [15]. The CNN layer takes the input data and conducts convolution operations on all the feature maps within it to discover high-level characteristics associated with the nature of the threat or malicious activity in the input data [14].

The proposed CNN-LSTM architecture consists of multiple layers designed to effectively capture both spatial and temporal features of network traffic data. Initially, the input data is passed through

convolutional layers with multiple filters to extract spatial feature representations. These layers are followed by activation functions (ReLU) and pooling layers to reduce dimensionality and retain important features.

The extracted feature maps are then flattened and passed to the LSTM layer, which is responsible for learning temporal dependencies and sequential patterns in the data. The LSTM network contains memory cells that retain information over time, enabling the detection of complex attack patterns that evolve across network traffic sequences.

Finally, the output from the LSTM layer is passed through fully connected (dense) layers, followed by a softmax classifier to categorize the traffic as either normal or malicious. This hybrid architecture enables the model to achieve improved accuracy by combining both spatial and temporal learning capabilities.

To compute the convolution between input and kernel data, the following equation must be utilized:

$$F(i,j)=(I*K)(i,j)$$

Where,

– I= input data; K= convolution kernel and F(I,j) = output feature map.

The CNN component uses multiple convolutional layers with kernel size 3×3 and max-pooling operations, while the LSTM layer consists of 64 memory units to capture temporal dependencies.

Once network traffic features have been extracted via convolution, they may then be passed to an LSTM network for the discovery of sequential dependencies amongst them. The LSTM model is capable of determining the time-based (temporal) pattern of occurrences during a cyberattack [2], [19].

E. Data Augmentation Using GAN

The Generative Adversarial Networks (GAN) used to generate synthetic attack samples for data augmentation enhances the performance of an Intrusion Detection System (IDS) [1], [15]. GANs use of deep learning allows for the generation of new data that is substantially similar to existing data sets (real data) by combining two neural networks, the Generator and the Discriminator.

Generator creates synthetic (fake) samples (xfake) from random input (z):

$$xfake = G(z)$$

where (z) is a random input and G(z) is a synthetic sample that corresponds to (z).

The Discriminator (D) used to determine whether the sample (x) is real or fake: D(x)

where D(x) is the probability (between 0 and 1) that the input data sample (x) is real.

The objective function for GANs is defined as follows:

$$\min G \max D\ V(D, G) = E[\log D(x)] + E[\log(1 - D(G(z)))]$$ Where,

G = Generator, D = Discriminator, x = real data, z = random noise,

D(x) = amount of data that is real

G(z) = fake data generated from noise

F. *Explainable Artificial Intelligence(XAI)*

Deep learning models are commonly viewed as being operated by black-box systems which makes it difficult for people to interpret how they make decisions[13], [18]. To remedy this situation, XAI techniques were applied to the output of the model to determine how it arrived at its prediction. The analysis consists of three parts:

1. Layer Analysis - This accounts for the contribution of each layer in the neural network using the attack detection model,

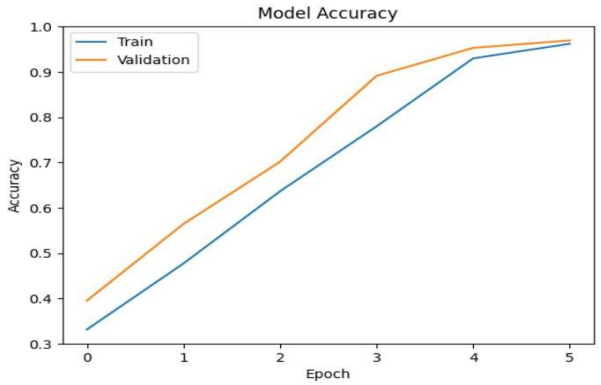
2. Feature Importance - This indicates the features in the network which had the biggest contribution on how the system makes its predictions,

3. Top Contributing Features - This indicates which features are most critical for identifying malicious network traffic.

IV. Results and Discussions

A. *Experimental Setup*

This study utilizes the Edge-IIoT Balanced Subset for Intrusion Detection dataset, which includes normal and malicious network traffic samples from Industrial Internet of Things (IIoT) environments and various types of cyberattacks and normal communication behavior. Prior to model training, the dataset was preprocessed to remove features that did not contribute to model prediction and to ensure that it was formatted appropriately for deep learning analysis [18]. The preprocessed dataset was then divided into training and testing datasets, the testing dataset contained the minority of records while the majority comprised the training dataset. The model was trained using deep learning during multiple epochs until the learning performance reached stability. During model training, the model learned patterns which were used to distinguish between normal network activity and malicious activity. Performance metrics of the system were evaluated with respect to traditional classification metrics that are utilized in research on intrusion detection systems through the use of the Edge-IIoT Balanced Subset for Intrusion Detection dataset [14], [17].



B. *Training Performance Analysis*

Fig. 2. Accuracy Graph

The above graph shown in Fig. 2. demonstrates how the proposed model's prediction capabilities improve with training. As training epochs are introduced, the model's prediction accuracy grows increases, indicating that the model is successfully extracting characteristics from the input data.

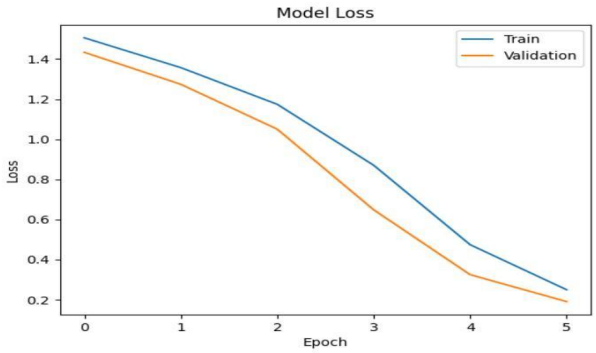


Fig. 3. Loss Graph

The above graph shown in Fig. 3. depicts errors between predicted outputs and actual labels during training. Since there was not an existing pattern of the input data at the beginning of training, the model produced a higher loss than at any later stage. As the model trains, the loss value should decrease with each subsequent epoch, indicating that the model has developed better prediction capabilities and reduced classification errors. Hence, the behavior of increased accuracy and decreased loss validates that the proposed model has learned characteristics of network traffic and improved its detection capabilities.

C. *Confusion Matrix Analysis*

A confusion matrix is an effective method of visualizing how well the classifier has classified instances into one of the five categories: "normal traffic," or "attack traffic". For the classification of each of the five classes, the matrix provides visually detailed information on both how many total samples were classified correctly and incorrectly into each category.

The diagonal elements of your confusion matrix correspond to the samples that were predicted correctly, meaning both "normal" and "attack" classified instances are shown as correctly classified instances for each attack type.

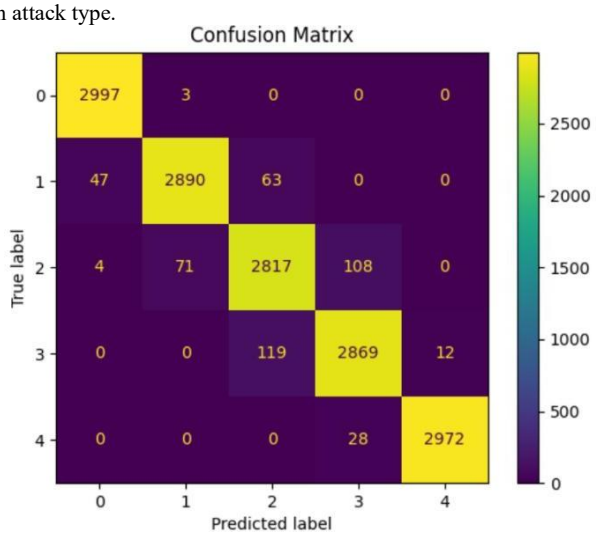


Fig. 4. Confusion Matrix

Similarly, the off-diagonal elements correspond to the instances of each attack type that were misclassified. A "good" performing intrusion detection system had a majority of the numbers in each category consisting primarily of diagonal elements (poorly categorized predictions consist primarily of off-diagonal elements).

D. *Explainability Analysis Using XAI*

XAI Techniques that Improve transparency and trust in an Intrusion Detection System applied for transparency and trust in an intrusion detection system, Explainable Artificial Intelligence (XAI) methods are used to analyze the prediction process of the deep learning model [13], [18].

1. Layer-wise Analysis - The layer-wise analysis provides insight regarding how the different layers of the neural network are utilized to process the input to the system. This structure of learning in layers gives the neural network the ability to learn and provide insights regarding very fine differences in the behaviors of normal and malicious actors.
2. Feature Importance - The feature importance analysis provides insight regarding the most important features of the input used to obtain prediction results. In some cases, certain network traffic features are found to have greater influence on the detection process, thereby identifying which features are most significant for identifying abnormal patterns [13], [18].
3. Top Contributing Features - The top contributing features highlighted during the analysis indicate the features that have the largest impact on the detection of an intrusion. That is, these features would provide valuable insight into which characteristics of network traffic are most helpful in discriminating between attacks and normal network communications.

E. *Final Results Summary*

The proposed intrusion detection model was tested using key classification metrics or performance metrics to assess its effectiveness in distinguishing between normal and harmful network traffic. These measurements offer a clear picture of how well the model performs in identifying complex patterns within cyber-physical systems.

Formulas:

Accuracy = (TP+TN) / (TP+TN +FP+ FN)

Precision = (TP) / (TP + FP Recall

= (TP) / (TP + FN)

F1-score = (2 × (Precision ×Recall)) /(Precision + Recall) Where,

TP = True Positive, TN = True Negative, FP = False Positive, FN = False Negative.

Model	Dataset	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
CNN Model	CPS Dataset	89.5	88.7	89	88.8
LSTM Model	CPS Dataset	91.2	90.5	91	90.7
CNN-LSTM Hybrid	Edge-IIoT	94.3	93.8	94.1	93.9
Proposed Model (CNN-LSTM + GAN + XAI)	Edge-IIoT	96.2	95.8	96.5	96.1

Table 2. Results Summary

Table 2 shows that the model consistently performs well across all metrics. It achieves 96.2% accuracy, indicating that it can generally classify input data correctly. With a precision of 95.8%, the model effectively reduces false positives. The recall rate of 96.5% demonstrates its strong capability in detecting true attack instances. The F1-score stands at 96.1%, signifying a good balance between precision and recall. These findings suggest that the model is both dependable and effective for use in intrusion detection within cyber-physical environments.

F. *Discussions*

The information obtained demonstrates that our model is able to find intrusion pattern activity within the Edge-IIoT Balanced Subset for Intrusion Detection dataset with high reliability in classifying both normal and malicious traffic, as demonstrated by the majority of prediction classes within the confusion matrix being accurate [14], [17]. This supports the model's ability to learn typical traffic patterns associated with IIoT-based cyberattacks [5], [20].

Using XAI analysis also helped to further understand the model by revealing what features contributed the most to influence the predicted outcome [13], [18]. Thus, our proposed method is both effective for identifying intrusions and more interpretable for use in cybersecurity applications than traditional approaches [13], [19]. Overall, this indicates that our proposed model can be effectively used for intrusion detection in both cyber-physical and IIoT environments [5], [7].

G. **Conclusion and Future Work**

This paper proposed a hybrid deep learning-based intrusion detection system for Cyber-Physical Systems using CNN, LSTM, and GAN with explainable AI techniques. The proposed model effectively captures both spatial and temporal patterns in network traffic data and improves detection accuracy. Experimental results on the Edge-IIoT dataset demonstrate that the model achieves high accuracy (96.2%) and provides interpretable insights into cyberattack detection. The integration of XAI techniques enhances transparency and helps in understanding the decision-making process of the model. This enhances the reliability and practical applicability of the system in real-world Cyber-Physical System environments. In future work, the model can be extended to handle real-time intrusion detection and tested on more diverse datasets. Additionally, optimization techniques can be applied to reduce computational complexity and improve scalability.

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